

Newman-Penrose Formalism Calculations

Stephani et al., Equation (31.60)

Metric

$$ds^2 = (2az^2 - 2b) dt \otimes dt + \left(-\frac{2}{2e^2x^6 - cx^5 - ax^4} \right) dx \otimes dx + (-4e^2x^4 + 2cx^3 + 2ax^2) dy \otimes dy + \left(-\frac{2}{az^2 - b} \right) dz \otimes dz$$

Coordinates

$$(x^\mu) = (t, x, y, z)$$

Null Tetrad

Covariant components

$$l = \left(\sqrt{-az^2 + b} \right) dt + \left(-\frac{1}{\sqrt{-az^2 + b}} \right) dz$$

$$n = \left(\sqrt{-az^2 + b} \right) dt + \left(\frac{1}{\sqrt{-az^2 + b}} \right) dz$$

$$m = \left(\frac{1}{\sqrt{-(2e^2x^2 - cx - a)x^2}} \right) dx + \left(i \sqrt{-(2e^2x^2 - cx - a)x^2} \right) dy$$

$$\bar{m} = \left(\frac{1}{\sqrt{-(2e^2x^2 - cx - a)x^2}} \right) dx + \left(-i \sqrt{-(2e^2x^2 - cx - a)x^2} \right) dy$$

Contravariant components

$$l = \left(\frac{\sqrt{-az^2 + b}}{2(az^2 - b)} \right) dt + \left(\frac{az^2 - b}{2\sqrt{-az^2 + b}} \right) dz$$

$$n = \left(\frac{\sqrt{-az^2 + b}}{2(az^2 - b)} \right) dt + \left(\frac{1}{2} \sqrt{-az^2 + b} \right) dz$$

$$m = \left(-\frac{(2e^2x^2 - cx - a)x^2}{2\sqrt{-2e^2x^2 + cx + a}} \right) dx + \left(-\frac{i\sqrt{-2e^2x^2 + cx + a}}{2(2e^2x^2 - cx - a)x} \right) dy$$

$$\bar{m} = \left(-\frac{(2e^2x^2 - cx - a)x^2}{2\sqrt{-2e^2x^2 + cx + a}} \right) dx + \left(\frac{i\sqrt{-2e^2x^2 + cx + a}}{2(2e^2x^2 - cx - a)x} \right) dy$$

Directional Derivatives

$$D = \frac{\sqrt{-az^2 + b}}{2(az^2 - b)} \partial_t + \frac{az^2 - b}{2\sqrt{-az^2 + b}} \partial_z$$

$$\Delta = \frac{\sqrt{-az^2 + b}}{2(az^2 - b)} \partial_t + \frac{1}{2} \sqrt{-az^2 + b} \partial_z$$

$$\delta = -\frac{(2e^2x^2 - cx - a)x^2}{2\sqrt{-2e^2x^2 + cx + a}} \partial_x - \frac{i\sqrt{-2e^2x^2 + cx + a}}{2(2e^2x^2 - cx - a)x} \partial_y$$

$$\bar{\delta} = -\frac{(2e^2x^2 - cx - a)x^2}{2\sqrt{-2e^2x^2 + cx + a}} \partial_x + \frac{i\sqrt{-2e^2x^2 + cx + a}}{2(2e^2x^2 - cx - a)x} \partial_y$$

Spin Coefficients

$$\rho = 0$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = 0$$

$$\lambda = 0$$

$$\nu = 0$$

$$\alpha = \frac{(8e^2x^2 - 3cx - 2a)x}{8\sqrt{-2e^2x^2 + cx + a}}$$

$$\beta = -\frac{(8e^2x^2 - 3cx - 2a)x}{8\sqrt{-2e^2x^2 + cx + a}}$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = \frac{az}{4\sqrt{-az^2 + b}}$$

$$\gamma = \frac{az}{4\sqrt{-az^2 + b}}$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = 2\,e^2x^4 - \frac{9}{16}cx^3 - \frac{1}{4}ax^2 - \frac{1}{8}a$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = 0$$

$$\Lambda = \frac{2}{3}e^2x^4 - \frac{3}{16}cx^3 - \frac{1}{12}ax^2 + \frac{1}{24}a$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = -\frac{4}{3}e^2x^4 + \frac{3}{8}cx^3 + \frac{1}{6}ax^2 - \frac{1}{12}a$$

$$\Psi_3 = 0$$

$$\Psi_4 = 0$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "*D*"